# INTRUSION DETECTION SYSTEMS

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# COURSE OBJECTIVES

- To introduce basic concepts of intrusion detection system.
- To understand Intrusion Prevention Systems, Network IDs protocol and model for intrusion analysis.
- To Understand when, where, how, and why to apply Intrusion Detection tools and techniques in order to improve the security posture of an enterprise.
- To learn agent development for intrusion detection and architectural models of IDs and IPs.

# What is an Intrusion Detection System?

- An intrusion detection system (IDS) is a network security tool that monitors network traffic and devices for known malicious activity, suspicious activity or security policy violations. IDSs can be software applications that are installed on endpoints or dedicated hardware devices that are connected to the network. Some IDS solutions are available as cloud services.



# HISTRORY OF INTRUSION DETECTION SYSTEMS

- system administrators performed intrusion detection by sitting in front of a console and monitoring user activities. They might detect intrusions by noticing, for example, that a vacationing user is logged in locally or that a seldom-used printer is unusually active. Although effective enough at the time, this early form of intrusion detection was ad hoc and not scalable.
- The next step in intrusion detection involved audit logs, which system administrators reviewed for evidence of unusual or malicious behavior. In the late '70s and early '80s, administrators typically printed audit logs on fan-folded paper, which were often stacked four- to five-feet high by the end of an average week.

- As storage became cheaper, audit logs moved online and researchers developed programs to analyze the data. However, analysis was slow and often computationally intensive and therefore intrusion detection programs were usually run at night when the system's user load was low. Therefore, most intrusions were still detected after they occurred.
- In the early '90s, researchers developed real-time intrusion detection systems that reviewed audit data as it was produced. This enabled the detection of attacks and attempted attacks as they occurred, which in turn allowed for real-time response, and, in some cases, attack preemption.

# Intrusion Detection Overview:

- The goal of intrusion detection is seemingly simple: to detect intrusions. However, the task is difficult, and in fact intrusion detection systems do not detect intrusions at all—they only identify evidence of intrusions, either while they're in progress or after the fact.
- Such evidence is sometimes referred to as an attack's "manifestation." If there is no manifestation, if the manifestation lacks sufficient information, or if the information it contains is untrustworthy, then the system cannot detect the intrusion.
- For example, suppose a house monitoring system is analyzing camera output that shows a person fiddling with the front door. The camera's video data is the manifestation of the occurring intrusion. If the camera lens is dirty or out of focus, the system will be unable to determine whether the person is a burglar or the owner.

# Detection techniques

- There are two basic categories of intrusion detection techniques: **anomaly detection** and **misuse detection**.
- **Anomaly detection**: Anomaly detection uses models of the intended behavior of users and applications, interpreting deviations from this "normal" behavior as a problem.
- A basic assumption of anomaly detection is that attacks differ from normal behavior. For example, we can model certain users' daily activity (type and amount) quite precisely. Suppose a particular user typically logs in around 10 a.m., reads mail, performs database transactions, takes a break between noon and 1 p.m., has very few file access errors, and so on. If the system notices that this same user logs in at 3 a.m., starts using compilers and debugging tools, and has numerous file access errors, it will flag this activity as suspicious.

- The main advantage of anomaly detection systems is that they can detect previously unknown attacks. By defining what's normal, they can identify any violation, whether it is part of the threat model or not. In actual systems, however, the advantage of detecting previously unknown attacks is paid for in terms of high false positive rates. Anomaly detection systems are also difficult to train in highly dynamic environments.

# Misuse detection

- Misuse detection systems essentially define what's wrong. They contain attack descriptions (or "signatures") and match them against the audit data stream, looking for evidence of known attacks. One such attack, for example, would occur if someone created a symbolic link to a UNIX system's password file and executed a privileged application that accesses the symbolic link. In this example, the attack exploits the lack of file access checks. The main advantage of misuse detection systems is that they focus analysis on the audit data and typically produce few false positives. The main disadvantage of misuse detection systems is that they can detect only known attacks for which they have a defined signature. As new attacks are discovered, developers must model and add them to the signature database.

# Signature-based detection

- Signature-based detection analyzes network packets for attack signatures, unique characteristics or behaviors that are associated with a specific threat. A sequence of code that appears in a particular malware variant is an example of an attack signature.
- A signature-based IDS maintains a database of attack signatures against which it compares network packets. If a packet triggers a match to one of the signatures, the IDS flags it.

# Different types of Intrusion Detection Systems

- Different types of Intrusion Detection systems are classified on the basis of different techniques and methods.
- **Network Intrusion Detection System (NIDS)** Network Intrusion Detection System sets up across the network at a specific planned point. NIDS monitors the traffic on the network from all devices. Similarly, it examines the traffic passing on the entire subnet and verifies it with the packet metadata and content. If NIDS detects any intrusion in the network, a warning alert is sent to the admin of that network.

- **Host Intrusion Detection System (HIDS)** Organizations install a Host Intrusion Detection System (HIDS) on independent networked devices. HIDS examines the incoming and outgoing traffic of the device only. It detects suspicious activities on the device and alerts the administrator. HIDS also checks whether system files are misplaced or not, for that it takes the screen capture of the current file system and verifies it with the screen capture of the previous file system.